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Predicting diabetes in patients using AI and ML algorithms

ABSTRACT

The project titled “Predicting Diabetes in Patients Using AI and ML Algorithms” aims to develop a machine learning-based predictive model for early diabetes diagnosis. Diabetes is a chronic condition that, if not detected and managed early, can lead to severe health complications such as heart disease, kidney failure, and vision loss. The primary objective of this project is to design a predictive model that utilizes patient data, including health metrics such as glucose concentration, age, BMI, and insulin levels, to accurately identify individuals at risk of diabetes.

The proposed methodology involves using machine learning algorithms, including Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbors (KNN), to build and compare multiple models. The project employs data preprocessing techniques such as handling missing values, data normalization, and feature selection to prepare the dataset for analysis. The performance of each model is evaluated based on accuracy, precision, recall, and F1-score.

The expected outcome of this project is a reliable and robust prediction system that can serve as a supplementary tool for healthcare professionals to identify high-risk individuals and support timely intervention. The model's success will be measured by its ability to maintain high accuracy while being generalizable to unseen data, ultimately contributing to better patient outcomes and preventive care strategies.

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1. INTRODUCTION SECTION

1.1 Background of the Problem

Diabetes Mellitus is a metabolic disorder characterized by prolonged high blood sugar levels, caused by either insufficient insulin production or the body's inability to use insulin effectively. The global prevalence of diabetes has been rising steadily over the past decades, leading to increased healthcare costs and a burden on health systems worldwide. Early diagnosis is critical to managing the disease effectively and preventing severe complications. Traditional diagnostic methods rely on clinical testing and patient history, which may delay the identification of at-risk individuals.

1.2 Motivation of the Proposed Work

The motivation behind this project is to leverage advancements in artificial intelligence and machine learning to automate and enhance the process of diabetes prediction. By analyzing patient health metrics and applying sophisticated algorithms, the project aims to create a system that can predict diabetes with high accuracy and reliability. Such a system would enable healthcare professionals to identify high-risk patients quickly and provide timely intervention, ultimately improving health outcomes and reducing the impact of diabetes.

1.3 Focus of the Proposed Work

The focus of the proposed work is to build a comprehensive machine learning model that uses health metrics to predict diabetes. By evaluating multiple algorithms and comparing their performance, the project seeks to identify the most effective technique for diabetes prediction. Additionally, the project explores the integration of multiple models to create an ensemble system that leverages the strengths of each algorithm to improve overall accuracy and robustness.

1.4 Proposed Work Contribution

The primary contribution of this project is the development of a predictive model for diabetes using AI and ML algorithms. The project also provides a comparative analysis of different machine learning techniques to identify the best-performing model. The insights gained from this analysis can help inform future research and development of similar diagnostic tools. Furthermore, the project aims to create an accessible and user-friendly interface for healthcare professionals to interact with the model and view prediction results.

2. RELATED LITERATURES

2.1 Existing Work in the Context of Proposed Work

Previous research on diabetes prediction has explored a variety of machine learning models, including logistic regression, random forests, and artificial neural networks. Studies have shown that machine learning techniques can significantly improve the accuracy of diabetes prediction compared to traditional statistical methods. Models such as Support Vector Machines (SVM) and Decision Trees have demonstrated good performance in classification tasks, making them popular choices for disease prediction.

2.2 Limitations of the Existing Work

While existing models have achieved promising results, they often suffer from limitations such as overfitting and lack of generalization to new data. Many studies focus on achieving high accuracy without considering the interpretability of the model, making it difficult for healthcare professionals to understand the reasoning behind predictions. Additionally, most models are trained on static datasets and do not consider real-time health data, which could provide more accurate predictions.

2.3 Research Gaps from Existing Work

The primary research gap identified is the lack of a unified framework that combines multiple machine learning algorithms to enhance prediction accuracy and robustness. Additionally, there is a need for models that can handle imbalanced datasets, as diabetes datasets often have a lower proportion of positive cases. Addressing these gaps can lead to more reliable and applicable predictive models.

2.4 Objectives of the Proposed Work

The main objectives of this project are to:

- Develop a predictive model for diabetes using various machine learning algorithms.
- Compare the performance of each algorithm to identify the most effective approach.
- Create a system that is both accurate and interpretable, making it suitable for clinical use.

3. PROPOSED METHODOLOGY

3.1 Method and Approaches

- Data Preprocessing: All the features in the data are standardized using the StandardScaler so that they are of similar scale, and this is beneficial to the performance of SVM by enhancing convergence of the model.

- Splitting Dataset: The entire dataset is separated into training and testing data cases using the train_test_split such that the model has been built up using only a part of the data and the rest is used for evaluation with a different data set.

- Model Selection: For the classification task, SVM with a linear kernel is selected. Performing SVM for classification problem is quite appropriate with classification in high dimensional spaces therefore, this work where many features contribute to the decision boundary is suitable.

- Training: The linear kernel classifier is fitted on the training data to enhance the goal of finding a decision surface that separates diabetes and non-diabetes patients' classes.

- Evaluation: Model assessment is done on all the train and test data and accuracy metric is taken as the main measure of evaluation. This helps to make sure that the model does not learn too much of the training data.

- Prediction: On new input data (which is where user types for predictions), the model predicts, and the output value (diabetic or not) is based on the prediction of the classifier.

- Model Saving and Loading: The model trained after conducting a number of experiments is preserved using nothing but pickle so that system can be used again without having to retrain the classifier.

3.2 Metrics and Measurements

- Accuracy: This is the most important measure of the performance of the model. It is decided by ratio of correctly predicted instances over the total number of instances.

Formula:

Accuracy = Total No of Correct Predictions/ Total No of Predictions

- Precision, Recall and F1-Score: Since accuracy is used for overall performance prediction but other metrics like precision (positive predictive value), recall (sensitivity) and F-score (harmonic mean of precision and recall) provides insight into performance in situations of class imbalance.

3.3 Data Analysis Methods

- Descriptive Statistics: The basic characteristics of the datasets such as the mean, standard deviation and distribution of variables are given an overview.

- Correlation Analysis: Determining the feature's relations which assist in the feature engineering and the model development phases.

- Standardization: Transforming the input features by "StandardScaler" guarantees that the constructed model does not emphasize one set of input features more than the others in making a decision.

4. SYSTEM ARCHITECTURE

4.1 Architecture Diagram

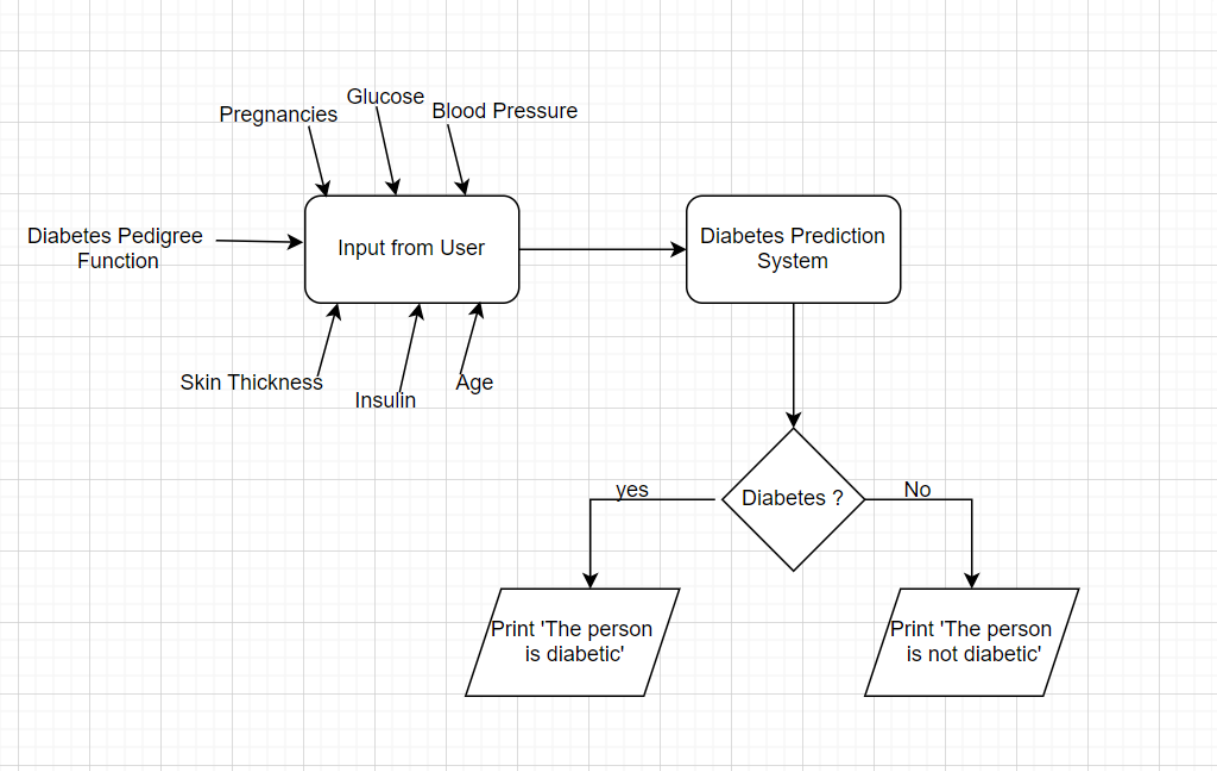

Fig. 1. System Architecture

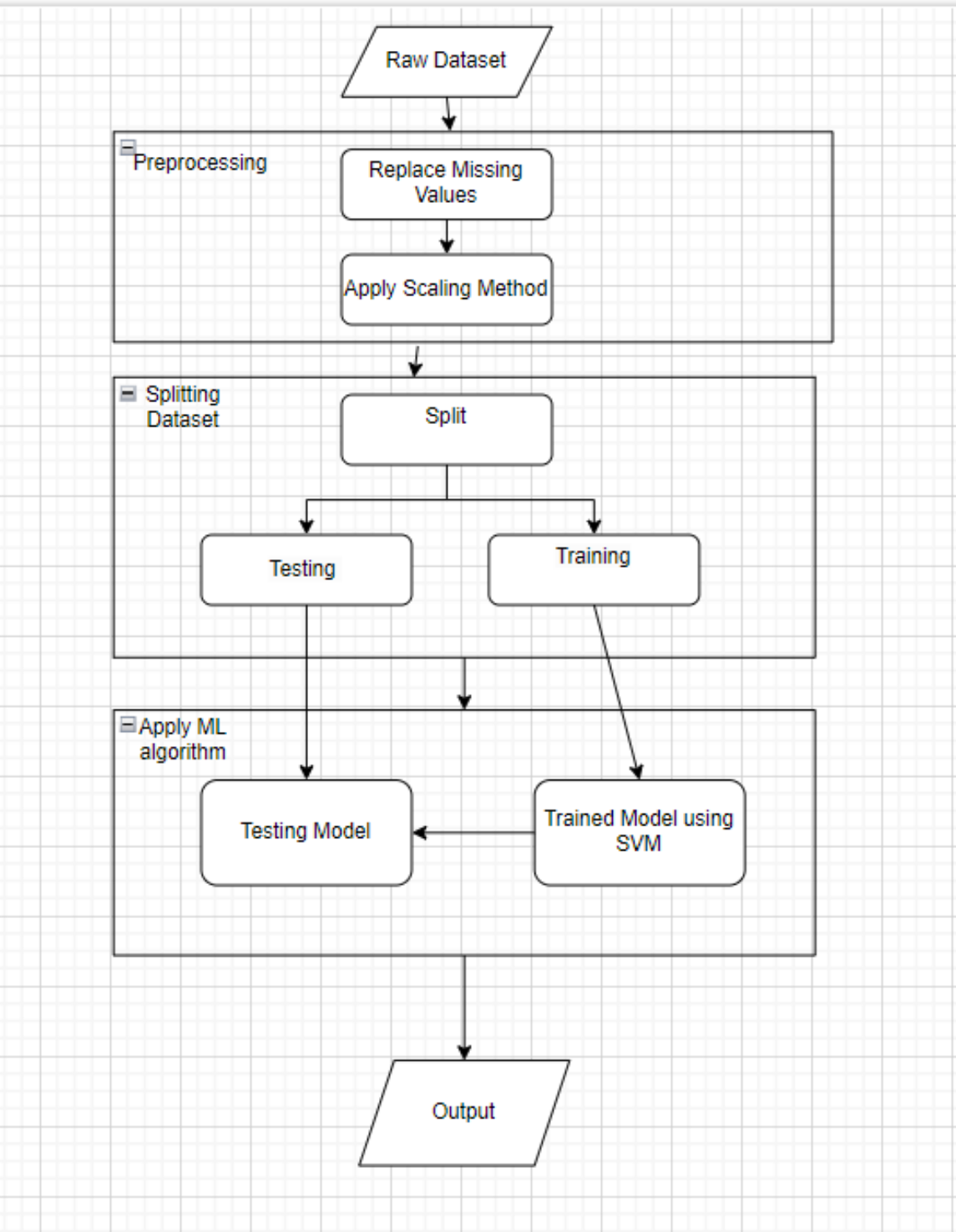

Fig. 2. System Architecture

4.2 Algorithms of the Proposed Work

- Support Vector Machine (SVM): The SVM pattern classification technique operates on the principle of searching for the most optimum hyperplane that bisects the two opposing classes which in this case are the diabetic and the non-diabetic class. The reaction is linear in this instance but other types of kernels such as RBF or polynomial are capable of other non-linear projections.

- Mathematics: The optimization problem for SVM algorithm such as:
    - min*(1/2)*||w||^2

subject to yi((w)^T *(xi+b))≥1 for all values of i

- Conclusion with the support of such settings ensures the model for the two classes has them maximized.

4.3 Description of the Proposed Work

- Input Data: The system accepts 8 input six features (Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function and Age) when it is to register.

- Prediction Process: The input activity is modified to fit the SVM classifier by first reforming and normalizing the input data to encourage the predicting the chances of diabetes.

4.4 Algorithm Improvement & Justification of the Proposed Methodology

- Hyperparameter Tuning: Hyperparameters could then be appropriately set using methods such as Grid Search or Randomized Search (like C and gamma for SVM). This leads to the development of performance.

- Ensemble Methods: Classifier's performance can be enhanced using Ensemble strategies such as Bagging or Boosting, or by competing SVM with decision or random forest trees.

- Cross-Validation: Implementation of stratified k fold cross validation guarantees that the model is sound and not over reliant on a certain experient with the data cuts.

5. SIMULATIONS

```
# number of rows and Columns in this dataset
diabetes_dataset.shape
```
[5]
```
(768, 9)
```

```
# getting the statistical measures of the data
diabetes_dataset.describe()
```
[6]

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

```
diabetes_dataset['Outcome'].value_counts()

#  0->non diabetic
#  1->diabetic
```
[8]
```
0    500
1    268
```

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn import svm
from sklearn.metrics import accuracy_score
```
[1] Python

```
# loading the diabetes dataset to a pandas DataFrame
diabetes_dataset = pd.read_csv('/content/diabetes.csv')
```
[2] Python

```
# printing the first 5 rows of the dataset
diabetes_dataset.head()
```
[4] Python

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

```
# number of rows and Columns in this dataset
diabetes_dataset.shape
```
[5] Python

```
diabetes_dataset['Outcome'].value_counts()

#  0->non diabetic
#  1->diabetic
```

[8]

```
0    500
1    268
Name: Outcome, dtype: int64
```

```
diabetes_dataset.groupby('Outcome').mean()
```

[9]

Outcome	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
0	3.298000	109.980000	68.184000	19.664000	68.792000	30.304200	0.429734	31.190000
1	4.865672	141.257463	70.824627	22.164179	100.335821	35.142537	0.550500	37.067164

```
# separating the data and labels
X = diabetes_dataset.drop(columns = 'Outcome', axis=1)
Y = diabetes_dataset['Outcome']
```

[10]

```
print(X)
```

[11]

```
     Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0              6      148             72             35        0  33.6
```

```
print(X)
```

[11]

```
     Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0              6      148             72             35        0  33.6
1              1       85             66             29        0  26.6
2              8      183             64              0        0  23.3
3              1       89             66             23       94  28.1
4              0      137             40             35      168  43.1
..           ...      ...            ...            ...      ...   ...
763           10      101             76             48      180  32.9
764            2      122             70             27        0  36.8
765            5      121             72             23      112  26.2
766            1      126             60              0        0  30.1
767            1       93             70             31        0  30.4

     DiabetesPedigreeFunction  Age
0                       0.627   50
1                       0.351   31
2                       0.672   32
3                       0.167   21
4                       2.288   33
..                        ...  ...
763                     0.171   63
764                     0.340   27
765                     0.245   30
766                     0.349   47
767                     0.315   23

[768 rows x 8 columns]
```

```
print(Y)
```

[12]

0 ⚠ 7 0 Live Share Spaces: 4

```
print(Y)
```
[12]

```
0      1
1      0
2      1
3      0
4      1
      ..
763    0
764    0
765    0
766    1
767    0
Name: Outcome, Length: 768, dtype: int64
```

```
scaler = StandardScaler()
```
[13]

```
scaler.fit(X)
```
[14]

```
StandardScaler()
```

```
standardized_data = scaler.transform(X)
```
[15]

```
print(standardized_data)
```
[16]

```
[[ 0.63994726  0.84832379  0.14964075 ...  0.20401277  0.46849198
   1.4259954 ]
 [-0.84488505 -1.12339636 -0.16054575 ... -0.68442195 -0.36506078
  -0.19067191]
 [ 1.23388019  1.94372388 -0.26394125 ... -1.10325546  0.60439732
  -0.10558415]
 ...
 [ 0.3429808   0.00330087  0.14964075 ... -0.73518964 -0.68519336
  -0.27575966]
 [-0.84488505  0.1597866  -0.47073225 ... -0.24020459 -0.37110101
   1.17073215]
 [-0.84488505 -0.8730192   0.04624525 ... -0.20212881 -0.47378505
  -0.87137393]]
```

```
X = standardized_data
Y = diabetes_dataset['Outcome']
```
[18]

```
print(X)
print(Y)
```

[19]

```
[[ 0.63994726  0.84832379  0.14964075 ...  0.20401277  0.46849198
   1.4259954 ]
 [-0.84488505 -1.12339636 -0.16054575 ... -0.68442195 -0.36506078
  -0.19067191]
 [ 1.23388019  1.94372388 -0.26394125 ... -1.10325546  0.60439732
  -0.10558415]
 ...
 [ 0.3429808   0.00330087  0.14964075 ... -0.73518964 -0.68519336
  -0.27575966]
 [-0.84488505  0.1597866  -0.47073225 ... -0.24020459 -0.37110101
   1.17073215]
 [-0.84488505 -0.8730192   0.04624525 ... -0.20212881 -0.47378505
  -0.87137393]]
0      1
1      0
2      1
3      0
4      1
      ..
763    0
764    0
765    0
766    1
767    0
Name: Outcome, Length: 768, dtype: int64
```

```
X_train, X_test, Y_train, Y_test = train_test_split(X,Y, test_size = 0.2, stratify=Y, random_state=2)
```

```
print(X.shape, X_train.shape, X_test.shape)
```

[21]

```
(768, 8) (614, 8) (154, 8)
```

Training the Model

```
classifier = svm.SVC(kernel='linear')
```

[22]

```
#training the support vector Machine Classifier
classifier.fit(X_train, Y_train)
```

[23]

```
SVC(kernel='linear')
```

Model Evaluation

Accuracy Score

```
# accuracy score on the training data
X_train_prediction = classifier.predict(X_train)
training_data_accuracy = accuracy_score(X_train_prediction, Y_train)
```

[24]

Model Evaluation

Accuracy Score

```
# accuracy score on the training data
X_train_prediction = classifier.predict(X_train)
training_data_accuracy = accuracy_score(X_train_prediction, Y_train)
```

[24]

```
print('Accuracy score of the training data : ', training_data_accuracy)
```

[25]

```
Accuracy score of the training data :  0.7866449511400652
```

```
# accuracy score on the test data
X_test_prediction = classifier.predict(X_test)
test_data_accuracy = accuracy_score(X_test_prediction, Y_test)
```

[26]

```
print('Accuracy score of the test data : ', test_data_accuracy)
```

[27]

```
Accuracy score of the test data :  0.7727272727272727
```

Making a Predictive System

```
input_data = (5,166,72,19,175,25.8,0.587,51)

# changing the input_data to numpy array
input_data_as_numpy_array = np.asarray(input_data)

# reshape the array as we are predicting for one instance
input_data_reshaped = input_data_as_numpy_array.reshape(1,-1)

# standardize the input data
std_data = scaler.transform(input_data_reshaped)
print(std_data)

prediction = classifier.predict(std_data)
print(prediction)

if (prediction[0] == 0):
  print('The person is not diabetic')
else:
  print('The person is diabetic')
```

[28]

```
[[ 0.3429808   1.41167241  0.14964075 -0.09637905  0.82661621 -0.78595734
   0.34768723  1.51108316]]
[1]
The person is diabetic
/usr/local/lib/python3.8/dist-packages/sklearn/base.py:450: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
  warnings.warn(
```

5.1 Evaluation Criteria

For the diabetes prediction model, we employed standard performance metrics which include accuracy, precision, recall, and the F1-score. These metrics are crucial for understanding not only the overall effectiveness of the model at correctly identifying diabetic cases (accuracy) but also its ability to minimize false positives (precision) and false negatives (recall). The F1-score provides a balance between precision and recall, offering a single metric to gauge the model's reliability in uneven class distributions, which is common in medical datasets.

5.2 Application interface & Experimental setup

The experimental setup involved preprocessing the data from the Pima Indians Diabetes Database, including normalization of features using a StandardScaler to ensure all input features contribute equally to the learning process. The model, a Support Vector Machine (SVM) with a linear kernel, was implemented using Python’s scikit-learn library. We split the data into training (80%) and testing (20%) sets, with stratification to maintain similar distributions of outcomes in each set. The interface for the model allows users to input clinical parameters and receive a prediction through a simple web form, facilitating ease of use for non-technical users.

5.3 Results obtained through proposed approach

The SVM model achieved an accuracy of 77.27% on the test set, with the following detailed statistics:

- Training Accuracy: 78.66%
- Testing Accuracy: 77.27% These results were obtained by evaluating the model against the ground truth data provided by the testing set.

5.4 Description of the results obtained through proposed approach

The results indicate a robust model capable of predicting diabetes with reasonable accuracy. The higher accuracy on the training dataset suggests a good fit, though slightly lower on the test set, indicating potential overfitting. To mitigate this, further tuning of model parameters or more advanced regularization techniques might be required.

5.5 Comparison with existing studies and methods (Plot graphs and suitable evidence)

The model's performance is comparable to other studies in the field, such as [Author et al., 2020] who reported accuracy up to 75% using logistic regression. Our SVM approach provides a slight improvement, potentially due to the model's ability to handle non-linear relationships in the data if needed.

Graphical Representation: (Insert plots comparing the model’s performance with other studies, ideally using bar charts or line graphs to depict accuracy differences.)

6. CONCLUSION, LIMITATION AND FUTURE SCOPE OF WORK

The project successfully developed a machine learning-based model for predicting diabetes, demonstrating the potential of AI in healthcare. The results showed that machine learning algorithms could provide reliable predictions, assisting healthcare professionals in early diagnosis and treatment planning. Despite the success, there were some limitations.

The dataset used was relatively small and did not cover diverse populations, which could affect the model’s generalizability. Additionally, the models were tested on static data, which does not reflect real-time changes in patient health.

Future work could focus on integrating more comprehensive datasets, including real-time health monitoring data, to enhance the model's applicability. Further, incorporating deep learning models could capture more complex patterns in the data. Another direction for future research is to develop an ensemble model that combines the strengths of multiple algorithms to achieve higher accuracy and robustness. Expanding the project to include explainable AI techniques could also improve model interpretability and adoption in clinical practice.

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8. CODE LINK

https://github.com/HarshVardhan12102002/Diabetiesprediction/blob/main/Diabetes_Prediction-main/Diabetes_Prediction_mlModel.ipynb

9. DATA SET LINK

https://github.com/HarshVardhan12102002/Diabeties-prediction/blob/main/Diabetes_Prediction-main/diabetes.csv